

# Management Summary - Bridge Program and AAISE Course Correlation v2

## Purpose

This artifact provides a management-level summary of the currently identified bridge-program direction and its emerging correlation to the future AAISE program.

It is intentionally scoped to the **gross changes currently identified** for review. It is not a final catalog proposal.

Its purpose is to show:

- the semester-by-semester developmental intent
- the current 10-152-nnn course structure
- the suggested bridge-program changes now under consideration
- the corresponding AAISE-facing nomenclature direction

## Program Reading

The bridge program is strongest when read as a four-semester developmental arc.

This summary keeps that arc visible while correlating the current bridge to a future AAISE naming structure.

## Semester 1 - Foundations and Controlled AI Entry (15 credits)

| Current Course                        | Credits | Suggested Bridge Direction                                        | Correlated AAISE Name                                 | Credits |
|---------------------------------------|---------|-------------------------------------------------------------------|-------------------------------------------------------|---------|
| 10-151-101 Introduction to Security   | 1       | Review later as needed; not part of the current gross change pass | n/a                                                   | n/a     |
| 10-152-117 Python Programming         | 2       | Keep and continue current AI-injected reframing                   | Foundations of Programming for AI-Augmented Engineers | 2       |
| 10-152-118 HTML, CSS & JavaScript     | 3       | Keep and continue AI-aware web foundations direction              | Web Foundations for AI-Augmented Engineers            | 3       |
| 10-152-119 Introduction to Algorithms | 3       | Keep and continue AI-aware logic and problem-solving direction    | Algorithmic Thinking for AI-Augmented Engineers       | 3       |
| 10-801-136 English Composition        | 3       | n/a                                                               | n/a                                                   | 3       |
| 10-801-189 Introductory Statistics    | 3       | n/a                                                               | n/a                                                   | 3       |

## Semester 2 - Data, Stack Depth, and Modern Development Reasoning (15/16 credits)

| Current Course                         | Credits | Suggested Bridge Direction                                                          | Correlated AAISE Name                              | Credits |
|----------------------------------------|---------|-------------------------------------------------------------------------------------|----------------------------------------------------|---------|
| 10-152-120 SQL Programming             | 3       | Keep practical SQL foundations in the bridge                                        | Data Query Foundations for AI-Augmented Engineers  | 3       |
| 10-152-121 Advanced Python Programming | 3       | Keep and expand around notebooks, libraries, ML workflow, and modern Python systems | Advanced Python Systems for AI-Augmented Engineers | 3       |

| Current Course                      | Credits | Suggested Bridge Direction                                                                         | Correlated AAISE Name                                          | Credits |
|-------------------------------------|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------|
| 10-152-122 C# Programming           | 3       | Rename in bridge to <b>C# &amp; ASP.NET Programming</b> and fold ASP.NET exposure into this course | Intelligent Application Development for AI-Augmented Engineers | 3       |
| 10-152-123 SQL Programming Advanced | 3       | Replace with <b>Modern Data Modeling for AI-Enabled Systems</b>                                    | Modern Data Modeling for AI-Enabled Systems                    | 3       |
| n/a                                 | n/a     | Possible bridge injection or pilot inside an existing 10-152-nnn course                            | Emerging Technology Intelligence for AI-Augmented Engineers    | 1       |
| 10-801-198 Speech                   | 3       | n/a                                                                                                | n/a                                                            | 3       |

## Semester 3 - AI-Injected Architecture and Applied System Context (16 credits)

| Current Course                                   | Credits | Suggested Bridge Direction                                                                                           | Correlated AAISE Name                                      | Credits |
|--------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------|
| 10-152-124 Introduction to Docker and Containers | 2       | Keep and reframe around packaging, deployment, and modern delivery literacy                                          | Containerized Application Deployment                       | 2       |
| 10-152-125 ASP.NET Programming                   | 3       | Drop as standalone ASP.NET course, merge core exposure into 10-152-122, and use this slot for the moved-in ML course | standalone ASP.NET removed from this point in the sequence | n/a     |
| 10-152-126 Cloud for Developers                  | 3       | Keep and expand around cloud context for AI-enabled systems                                                          | Cloud Platforms for AI-Enabled Systems                     | 3       |
| 10-152-127 AI-Driven Systems Analysis            | 3       | Keep and use as a primary location for <b>RBA injection and architectural framing</b>                                | AI-Driven Systems Analysis and Responsible Design          | 3       |
| 10-152-128 C++ Programming                       | 2       | Rename in bridge to <b>C++ Systems Performance and AI Runtime Foundations</b>                                        | Systems Performance and AI Runtime Foundations             | 2       |
| 10-152-131 Machine Learning and AI               | 3       | Move into Semester 3 as the machine-learning foundation before deep learning                                         | Machine Learning Foundations and Applications              | 3       |
| 10-809-195 Economics                             | 3       | n/a                                                                                                                  | n/a                                                        | 3       |

## Semester 4 - Delivery, Integration, and Demonstrated Capability (16 credits)

| Current Course                                        | Credits | Suggested Bridge Direction                                                                                                      | Correlated AAISE Name                                                  | Credits |
|-------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------|
| 10-152-129 Agile Practices                            | 3       | Keep and continue expansion into GitHub, review, and CI/CD workflow                                                             | Agile Delivery for AI-Enabled Teams                                    | 3       |
| 10-152-130 Software Career Experience                 | 2       | Retain current bridge nomenclature, but apply the <b>Applied Software Integration Studio</b> concept as the governing direction | Applied Software Integration Studio                                    | 2       |
| 10-152-131 Machine Learning and AI                    | 3       | Move earlier into Semester 3 so Semester 4 can carry the deeper follow-on course                                                | Machine Learning Foundations and Applications now occurs in Semester 3 | n/a     |
| 10-152-132 Mobile Development                         | 2       | Review later; Focus on Flutter/Dart                                                                                             | n/a                                                                    | 2       |
| 10-152-133 Enterprise Data Systems & API Architecture | 3       | Rename and repurpose this slot as the Semester 4 deep-learning follow-on course                                                 | Deep Learning Frameworks and Applications                              | 3       |
| n/a                                                   | n/a     | Possible bridge injection, pilot, or partial integration near the end of the bridge sequence                                    | Technical Intelligence Studio for AI-Augmented Engineers               | 3       |

| Current Course                           | Credits | Suggested Bridge Direction | Correlated AAISE Name | Credits |
|------------------------------------------|---------|----------------------------|-----------------------|---------|
| 10-809-199 Psychology of Human Relations | 3       | n/a                        | n/a                   | 3       |

## Visual Key

- Yellow = rename / reframed title
- Red = drop / merge / absorbed elsewhere
- Green = new / added course direction

## Gross Bridge Changes in This Pass

The current gross bridge changes under active consideration are:

- replace 10-152-123 SQL Programming Advanced with **Modern Data Modeling for AI-Enabled Systems**
- rename 10-152-122 C# Programming to **C# & ASP.NET Programming**
- drop 10-152-125 ASP.NET Programming as a standalone course, merge its core exposure into the revised C# course, and repurpose the slot
- rename 10-152-128 C++ Programming to **C++ Systems Performance and AI Runtime Foundations**
- use 10-152-127 AI-Driven Systems Analysis as a primary carrier for **RBA-based systems framing**
- move 10-152-131 Machine Learning and AI into **Semester 3**
- rename and repurpose 10-152-133 Enterprise Data Systems & API Architecture as **Deep Learning Frameworks and Applications**
- retain 10-152-130 Software Career Experience as the bridge nomenclature, while redirecting its underlying concept toward **Applied Software Integration Studio**

## New AAISE Additions and Injection Note

Two newly proposed AAISE-oriented courses remain relevant to the future program structure:

- **Emerging Technology Intelligence for AI-Augmented Engineers**
- **Technical Intelligence Studio for AI-Augmented Engineers**

These appear strongest as explicit AAISE courses.

For the bridge program, one, the other, or both may need to be injected into existing 10-152-*nnn* courses so the concepts can be tested under current governance constraints before a fuller AAISE rollout.

## Management-Level Read

This correlation suggests that the bridge does not need a total rebuild in order to become meaningfully more AI-era aligned.

Instead, the strongest near-term direction appears to be:

- preserve the semester progression
- modernize course naming where governance latitude allows
- use broader, adaptation-friendly course identity for the future AAISE
- fold redundant legacy content together where appropriate
- create clearer space for RBA, AI-injected systems reasoning, and applied integration

This keeps the bridge practical while making the future AAISE structure more legible.

## Review Boundary

This artifact reflects only the currently identified first-pass bridge and AAISE correlations requested for review.

Further adaptation and refactoring can be layered after management feedback on this gross-change pass.